1 Gain law

(bars denote normalized quantities: lengths by R; $a_o := \Delta t/(\gamma_N R)$ [1/pN] and $a_{\text{nom}} := \Delta t/\gamma_N = a_o R$ [μm/pN]; $\bar{a} := a/a_{\text{nom}}$; forces by a_o: $\bar{f} := a_o f$, so that $\bar{a}\,\bar{f} = \Delta\bar{w}$; $0 \le \bar{a} \le 1$, far field 1, contact 0; hats sit on the normalized quantities $\hat{\bar{a}}_w$, $\hat{b}$, $\delta\hat{\bar{w}}$; $e_\theta = \theta - \hat{\theta}$)

$$
\begin{aligned}
\bar{a}(\bar{w}, b) &= 1 - \frac{1}{b\,(\bar{w} - \bar{w}_s)} \\
\bar{a}'(\bar{w}, b) := \frac{\partial \bar{a}}{\partial \bar{w}} &= \frac{1}{b\,(\bar{w} - \bar{w}_s)^2} = b\big(1 - \bar{a}\big)^2 \\
\frac{\partial \bar{a}}{\partial b} &= \frac{1}{b^2(\bar{w} - \bar{w}_s)} = \frac{1 - \bar{a}}{b}
\end{aligned}
$$

2 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k - d + j - 1],\ j = 1, \ldots, d+1,\ d = 2\big)$

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta\bar{w}_d[k] \\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1 - \lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + b[k]\big(1 - \bar{a}_w[k]\big)^2\,\Delta\bar{w}[k] \\
b[k+1] &= b[k]
\end{aligned}
$$

$$
\begin{aligned}
F_{dw}[k] &= \bar{f}_{dw}[k] + (1 - \lambda_c)\sum_{i=1}^{d} \bar{f}_{dw}[k - i] \\
\varepsilon_w[k] &= \bar{a}_w[k]\,\bar{f}_T[k] + (1 - \lambda_c)\sum_{i=1}^{d} \bar{a}_w[k - i]\,\bar{f}_T[k - i] + (1 - \lambda_c)\,n_w[k]
\end{aligned}
$$

$$
\begin{aligned}
\bar{a}_w[k] &= \bar{a}\big(\bar{w}[k],\ b\big) \\
\hat{\bar{a}}_w[k] &= \bar{a}\big(\hat{\bar{w}}[k],\ \hat{b}[k]\big), \qquad \hat{\bar{w}}[k] = \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k] \\
\bar{w}[k] - \hat{\bar{w}}[k] &= -e_{\delta\bar{w}_3}[k] \\
e_{\bar{a}_w}[k] &= \frac{\partial \bar{a}}{\partial \bar{w}}\big(\bar{w}[k] - \hat{\bar{w}}[k]\big) + \frac{\partial \bar{a}}{\partial b}\,e_b[k] + \mathcal{O}\big(e^2\big) \\
&= -\hat{b}\big(1 - \hat{\bar{a}}_w\big)^2 e_{\delta\bar{w}_3}[k] + \frac{1 - \hat{\bar{a}}_w}{\hat{b}}\,e_b[k]
\end{aligned}
$$

$$\delta\bar{w}_3[k+1] = \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\Big(-\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 e_{\delta\bar{w}_3}[k] + \frac{1-\hat{\bar{a}}_w}{\hat{b}}\,e_b[k]\Big) - \varepsilon_w[k]$$

$$\begin{aligned}\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] \\ &\quad + F_{dw}[k]\Big(-\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 e_{\delta\bar{w}_3}[k] + \frac{1-\hat{\bar{a}}_w}{\hat{b}}\,e_b[k]\Big) + \varepsilon_w[k]\end{aligned}$$

3 Estimator

$$\mathbf{x} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ b\,\big]^\top \qquad (d=2)$$

$$\begin{aligned}\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\ \delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\ \delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\ \Delta\hat{\bar{w}}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k] \\ \hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2\,\Delta\hat{\bar{w}}[k] \\ \hat{b}[k+1] &= \hat{b}[k]\end{aligned}$$

4 Error dynamics

(True − Estimator; coefficients at the estimate)

$$\begin{aligned}e_{\Delta\bar{w}}[k] &= \Delta\bar{w}[k] - \Delta\hat{\bar{w}}[k] \\ &= \Big((1-\lambda_c) - F_{dw}[k]\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2\Big)e_{\delta\bar{w}_3}[k] + F_{dw}[k]\,\frac{1-\hat{\bar{a}}_w}{\hat{b}}\,e_b[k] + \varepsilon_w[k]\end{aligned}$$

$$b[k]\big(1-\bar{a}_w[k]\big)^2 = \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2 - 2\,\hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)\,e_{\bar{a}_w}[k] + \big(1-\hat{\bar{a}}_w[k]\big)^2 e_b[k] + \mathcal{O}\big(e^2\big)$$

$$\begin{aligned}e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\ e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\ e_{\delta\bar{w}_3}[k+1] &= \Big(\lambda_c + F_{dw}\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2\Big)e_{\delta\bar{w}_3}[k] - F_{dw}\,\frac{1-\hat{\bar{a}}_w}{\hat{b}}\,e_b[k] - \varepsilon_w[k] \\ e_{\bar{a}_w}[k+1] &= \Big[1 - 2\hat{b}\big(1-\hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\big)\Big]e_{\bar{a}_w}[k] \\ &\quad + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2\Big((1-\lambda_c) - F_{dw}\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2\Big)e_{\delta\bar{w}_3}[k] \\ &\quad + \Big(\big(1-\hat{\bar{a}}_w\big)^2\big(\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\big) + F_{dw}\big(1-\hat{\bar{a}}_w\big)^3\Big)e_b[k] + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2\varepsilon_w[k] \\ e_b[k+1] &= e_b[k]\end{aligned}$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c + F_{dw}\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 & 0 & -F_{dw}\,\dfrac{1-\hat{\bar{a}}_w}{\hat{b}} \\ 0 & 0 & \begin{array}{c}(1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 \\ -F_{dw}\,\hat{b}^2\big(1-\hat{\bar{a}}_w\big)^4\end{array} & \begin{array}{c}1-2\hat{b}\big(1-\hat{\bar{a}}_w\big) \\ \times\big(\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\big)\end{array} & \begin{array}{c}\big(1-\hat{\bar{a}}_w\big)^2 \\ \times\big(\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\big) \\ +F_{dw}\big(1-\hat{\bar{a}}_w\big)^3\end{array} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = \Big[\,0,\ 0,\ -\varepsilon_w,\ \hat{b}\big(1-\hat{\bar{a}}_w\big)^2\varepsilon_w,\ 0\,\Big]^\top$$

5 Measurement

$\big(\nabla_d\bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d]\big)$

$$y_1[k] = \delta\bar{w}_1[k] + n_w[k]$$
$$y_2[k] = \bar{a}_w[k-d] + n_a[k], \qquad \bar{a}_w[k-d] = \bar{a}_w[k] - b\big(1-\bar{a}_w[k]\big)^2\,\nabla_d\bar{w}_d[k]$$
$$\hat{y}_1[k] = \delta\hat{\bar{w}}_1[k]$$
$$\hat{y}_2[k] = \hat{\bar{a}}_w[k] - \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2\,\nabla_d\bar{w}_d[k]$$

$$e_{y_2}[k] = \frac{\partial\hat{y}_2[k]}{\partial\hat{\bar{a}}_w[k]}\,e_{\bar{a}_w}[k] + \frac{\partial\hat{y}_2[k]}{\partial\hat{b}[k]}\,e_b[k] + n_a[k]$$
$$\frac{\partial\hat{y}_2[k]}{\partial\hat{\bar{a}}_w[k]} = 1 + 2\,\hat{b}\big(1-\hat{\bar{a}}_w\big)\,\nabla_d\bar{w}_d[k], \qquad \frac{\partial\hat{y}_2[k]}{\partial\hat{b}[k]} = -\big(1-\hat{\bar{a}}_w\big)^2\,\nabla_d\bar{w}_d[k]$$
$$e_{\bar{a}_w}[k] = -\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 e_{\delta\bar{w}_3}[k] + \frac{1-\hat{\bar{a}}_w}{\hat{b}}\,e_b[k]$$
$$e_{y_2}[k] = -\Big(1 + 2\,\hat{b}\big(1-\hat{\bar{a}}_w\big)\,\nabla_d\bar{w}_d[k]\Big)\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 e_{\delta\bar{w}_3}[k] + \Big(\frac{1-\hat{\bar{a}}_w}{\hat{b}} + \big(1-\hat{\bar{a}}_w\big)^2\nabla_d\bar{w}_d[k]\Big)e_b[k] + n_a[k]$$

$$H_{23}[k] = -\Big(1 + 2\,\hat{b}\big(1-\hat{\bar{a}}_w\big)\,\nabla_d\bar{w}_d[k]\Big)\hat{b}\big(1-\hat{\bar{a}}_w\big)^2$$
$$H_{25}[k] = \frac{1-\hat{\bar{a}}_w}{\hat{b}} + \big(1-\hat{\bar{a}}_w\big)^2\nabla_d\bar{w}_d[k]$$
$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & H_{23}[k] & 0 & H_{25}[k] \end{bmatrix}$$
$$\mathbf{e}_y[k] = H[k]\,\mathbf{e}^-[k] + \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}$$

6 Noise

$$Q[k] = Q_{33}[k]\,\mathbf{g}[k]\,\mathbf{g}[k]^\top, \qquad \mathbf{g}[k] = \Big[\,0,\ 0,\ -1,\ \hat{b}[k]\big(1-\hat{\tilde{a}}_w[k]\big)^2,\ 0\,\Big]^\top, \qquad Q_{55} = 0$$

$$Q_{33}[k] = \kappa_T\Big(\hat{\tilde{a}}_w[k] + (1-\lambda_c)^2\big(\hat{\tilde{a}}_w[k-1] + \hat{\tilde{a}}_w[k-2]\big)\Big) + (1-\lambda_c)^2\sigma_{n_w}^2, \qquad \kappa_T = \frac{4k_BT\,a_o}{R}$$

$$R[k] = \mathrm{diag}\big(R_1,\ R_2[k]\big), \qquad R_1 = \sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2}, \qquad R_2[k] = 2a_{\mathrm{cov}}^2\,IF_{\mathrm{var}}\big(\hat{\tilde{a}}_w[k]\big)\big(\hat{\tilde{a}}_w[k]+\xi\big)^2$$